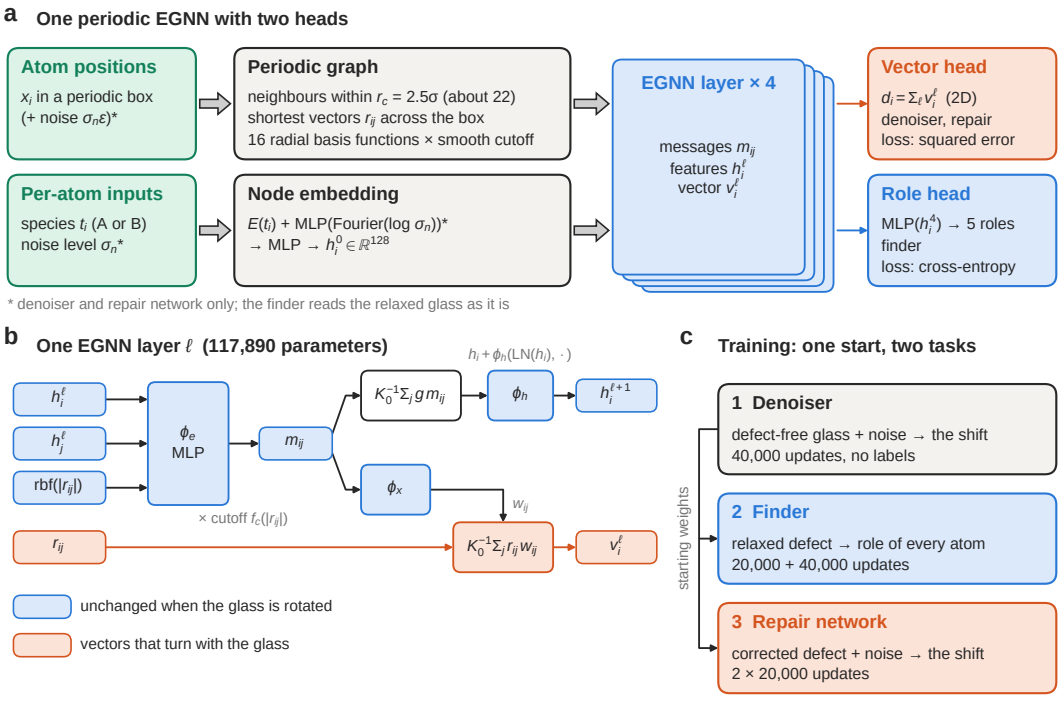
a
One periodic EGNN with two heads
Atom positions
x_i in a periodic box
(+ noise $\sigma_n\varepsilon$)*
Per-atom inputs
species t_i (A or B)
noise level σ_n*
Periodic graph
neighbours within r_c = 2.5σ (about 22)
shortest vectors r_{ij} across the box
16 radial basis functions × smooth cutoff
Node embedding
$E(t_i)$ + MLP(Fourier(log σ_n))*
→ MLP → $h_i^0 \in \mathbb{R}^{128}$
EGNN layer × 4
messages m_{ij}
features h_i^ℓ
vector v_i^ℓ
Vector head
$d_i = \Sigma_\ell v_i^\ell$ (2D)
denoiser, repair
loss: squared error
Role head
MLP(h_i^4) → 5 roles
finder
loss: cross-entropy
* denoiser and repair network only; the finder reads the relaxed glass as it is
b
One EGNN layer ℓ (117,890 parameters)
$h_i + \phi_h(\mathrm{LN}(h_i), \cdot)$
h_i^ℓ
h_j^ℓ
rbf($|r_{ij}|$)
ϕ_e MLP
× cutoff $f_c(|r_{ij}|)$
m_{ij}
$K_0^{-1}\Sigma_j g\, m_{ij}$
ϕ_h
$h_i^{\ell+1}$
ϕ_x
w_{ij}
r_{ij}
$K_0^{-1}\Sigma_j r_{ij} w_{ij}$
v_i^ℓ
unchanged when the glass is rotated
vectors that turn with the glass
c
Training: one start, two tasks
starting weights
1 Denoiser
defect-free glass + noise → the shift
40,000 updates, no labels
2 Finder
relaxed defect → role of every atom
20,000 + 40,000 updates
3 Repair network
corrected defect + noise → the shift
2 × 20,000 updates